

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

M2 Practical No: 5

Aim: Performing independent two-sample t-tests using `t.test()` with grouping (R).

Code:

1. Load Data

```
rm(list = ls())
```

```
data <- read.csv("song_data.csv")
```

```
data$final_total_points <- as.numeric(data$final_total_points)
data <- subset(data, gender %in% c("Female", "Male"))
data <- na.omit(data[c("final_total_points", "gender")])
```

2. Check structure

```
print("--- 2. Data Structure ---")
str(data)
```

3. Descriptive statistics (Using aggregate)

```
print("\n--- 3. Descriptive Statistics (by Gender) ---")
aggregate(final_total_points ~ gender, data = data,
  FUN = function(x) c(
    Mean = mean(x),
    SD = sd(x),
    N = length(x)
  ))
```

4. Normality Test (Shapiro-Wilk)

```
print("\n--- 4. Normality Test (Female) ---")
shapiro.test(data$final_total_points[data$gender == "Female"])
```

```
print("\n--- 4. Normality Test (Male) ---")
shapiro.test(data$final_total_points[data$gender == "Male"])
```

5. Equality of Variance Test (F-test/Variance Ratio Test)

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```
print("\n--- 5. Equality of Variance Test (F-test) ---")  
var.test(final_total_points ~ gender, data = data)
```

6. Independent Two Sample t-test

6a. Welch t-test (Unequal variance) - DEFAULT

```
print("\n--- 6a. Welch t-test (Unequal variance) ---")  
t_test_result <- t.test(final_total_points ~ gender, data = data)  
print(t_test_result)
```

6b. Equal variance assumed (Student's t-test)

```
print("\n--- 6b. Equal variance assumed (Student's t-test) ---")  
t_test_equal_var <- t.test(final_total_points ~ gender, data = data, var.equal = TRUE)  
print(t_test_equal_var)
```

7. Manual Method (Subsetting)

```
group_A <- data$final_total_points[data$gender == "Female"]  
group_B <- data$final_total_points[data$gender == "Male"]  
  
print("\n--- 7. Manual Method Check (Should match 6a) ---")  
t.test(group_A, group_B)
```

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Output:

The image displays two screenshots of the RStudio interface, showing the execution of R code and the resulting output in the console and environment panes.

Top Screenshot:

Console Output:

```
R - R4.5.2 - ~/...
> rm(list = ls())
> data <- read.csv("song_data.csv")
> data$final_total_points <- as.numeric(data$final_total_points)
> data <- subset(data, gender %in% c("Female", "Male"))
> data <- na.omit(data[c("final_total_points", "gender")])
> print("--- 2. Data Structure ---")
[1] "--- 2. Data Structure ---"
> str(data)
'data.frame':   331 obs. of  2 variables:
 $ final_total_points: num  268 30 59 123 92 362 96 583 129 526 ...
 $ gender             : chr  "Female" "Male" "Female" "Male" ...
- attr(*, "na.action")= 'omit' Named int [1:187] 2 4 6 12 15 17 21 22 25 26 ...
- attr(*, "names")= chr [1:187] "2" "4" "6" "12" ...
> print("--- 3. Descriptive Statistics (by Gender) ---")
[1] "\n--- 3. Descriptive Statistics (by Gender) ---"
> aggregate(final_total_points ~ gender, data = data,
+           FUN = function(x) c(
+             Mean = mean(x),
+             SD = sd(x),
+             N = length(x)
+           ))
  gender final_total_points.Mean final_total_points.SD final_total_points.N
1 Female          154.5060          134.1420          168.0000
2 Male           177.3129          156.2123          163.0000
> print("--- 4. Normality Test (Female) ---")
[1] "\n--- 4. Normality Test (Female) ---"
> shapiro.test(data$final_total_points[data$gender == "Female"])

Shapiro-wilk normality test

data:  data$final_total_points[data$gender == "Female"]
W = 0.87235, p-value = 9.13e-11

> print("--- 4. Normality Test (Male) ---")
[1] "\n--- 4. Normality Test (Male) ---"
> shapiro.test(data$final_total_points[data$gender == "Male"])

Shapiro-wilk normality test

data:  data$final_total_points[data$gender == "Male"]
W = 0.88544, p-value = 6.810e-10
```

Environment Pane:

Object	Class	Attributes
data	data.frame	331 obs. of 2 variables
t_test_equal_var	list	List of 10
t_test_result	list	List of 10

Bottom Screenshot:

Console Output:

```
R - R4.5.2 - ~/...
> print("--- 5. Equality of Variance Test (F-test) ---")
[1] "\n--- 5. Equality of Variance Test (F-test) ---"
> var.test(final_total_points ~ gender, data = data)

F test to compare two variances

data:  final_total_points by gender
F = 0.73739, num df = 167, denom df = 162, p-value = 0.0513
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.5424153 1.0017325
sample estimates:
ratio of variances
 0.7373932

> print("--- 6a. Welch t-test (Unequal variance) ---")
[1] "\n--- 6a. Welch t-test (Unequal variance) ---"
> t_test_result <- t.test(final_total_points ~ gender, data = data)
> print(t_test_result)

Welch Two Sample t-test

data:  final_total_points by gender
t = -1.4232, df = 318.55, p-value = 0.1557
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
 -54.336018  8.722156
sample estimates:
mean in group Female   mean in group Male
          154.5060           177.3129

> print("--- 6b. Equal variance assumed (Student's t-test) ---")
[1] "\n--- 6b. Equal variance assumed (Student's t-test) ---"
> t_test_equal_var <- t.test(final_total_points ~ gender, data = data, var.equal = TRUE)
> print(t_test_equal_var)

Two Sample t-test

data:  final_total_points by gender
t = -1.4264, df = 329, p-value = 0.1547
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
```

Environment Pane:

Object	Class	Attributes
data	data.frame	331 obs. of 2 variables
t_test_equal_var	list	List of 10
t_test_result	list	List of 10

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The screenshot shows the RStudio interface with the following content:

Source

```
R - R4.5.2 - ~/...
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
```

Console

```
95 percent confidence interval:
-54.336018  8.722156
sample estimates:
mean in group Female    mean in group Male
      154.5060           177.3129

> print("\n--- 6b. Equal variance assumed (Student's t-test) ---")
[1] "\n--- 6b. Equal variance assumed (Student's t-test) ---"
> t_test_equal_var <- t.test(final_total_points ~ gender, data = data, var.equal = TRUE)
> print(t_test_equal_var)

      Two Sample t-test

data: final_total_points by gender
t = -1.4264, df = 329, p-value = 0.1547
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
-54.260085  8.646223
sample estimates:
mean in group Female    mean in group Male
      154.5060           177.3129

> group_A <- data$final_total_points[data$gender == "Female"]
> group_B <- data$final_total_points[data$gender == "Male"]
> print("\n--- 7. Manual Method Check (Should match 6a) ---")
[1] "\n--- 7. Manual Method Check (Should match 6a) ---"
> t.test(group_A, group_B)

      Welch Two Sample t-test

data: group_A and group_B
t = -1.4232, df = 318.55, p-value = 0.1557
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-54.336018  8.722156
sample estimates:
mean of x mean of y
      154.5060      177.3129

> |
```

Environment

Object	Type	Value
data	data.frame	331 obs. of 2 variables
t_test_equal_var	htest	List of 10
t_test_result	htest	List of 10

Values

Object	Value
group_A	num [1:168] 268 59 362 583 129 122 168 93 ...
group_B	num [1:163] 30 123 92 96 526 182 126 78 15...

Files **Plots** **Packages** **Help** **Viewer** **Presentation**

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